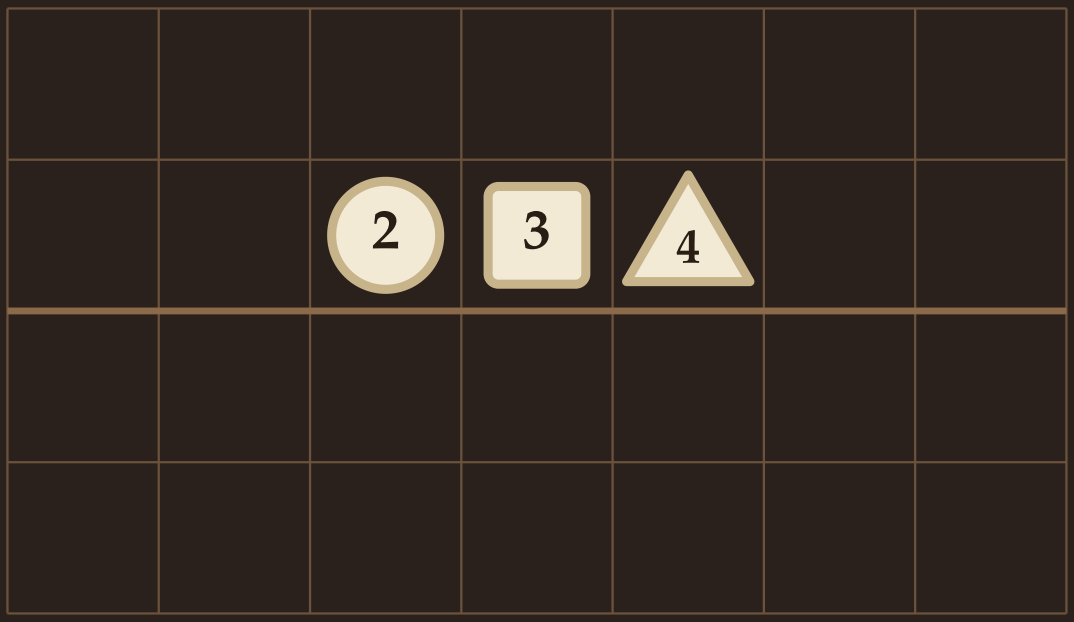
modern

RITHMOMACHIA

PLAYER'S MANUAL



TACTICS · TRIUMPH · STRATEGY · OPENINGS · ENDGAMES

with 28 puzzles and their full solutions

Modern Rithmomachia

Player's Manual

Rules · Tactics · Triumph · Strategy
Openings · Endgames · Puzzles

Modern Rithmomachia — Player's Manual

A game by Guillaume Enguehard. Revised edition, November 2026.

Reference rules: setup B, capture by moving, encounter between different shapes, assault by multiplication or division, ambush by sum or difference, victory by triumph, loss by exhaustion or by having no legal move, draw after 60 plies without a capture.

The principles, statistics and puzzles in this manual were established by analysing games between game engines, then confirmed by a stronger engine. Every game was checked by a second, independent implementation of the rules.

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Foreword

Modern Rithmomachia is a version of the medieval "battle of numbers" that can be played today. You win by lining up three of your pieces in the opponent's half to form a progression: 2-3-4, 2-4-6, 2-3-6 or 3-4-6. Pieces capture one another through relations between numbers: equality, multiplication, addition.

The game is new: it has neither masters nor theory. The principles in this manual therefore come from computer analysis, carried out in three stages.

1. **The exact foundations**: who can capture whom, where a triumph can form, how many lines pass through each square. These are facts of the rules, not opinions.
2. **Observation**: 500 games by a club-level engine, plus controlled experiments. The same position is replayed hundreds of times while changing a single element (one number fewer, a square in place of a triangle).
3. **Confirmation**: a much stronger engine, which beats the first one 39 games out of 40, replayed every experiment. Only the principles that held up appear here as principles. Those that were disproved are flagged as such, because knowing what does not work is also part of the theory.

The percentages quoted are scores from these experiments: "75%" means that the side in question scored 75 points out of 100, with a win counting 1 and a draw ½. They give the order of magnitude of an advantage, not a law. All the puzzles have been checked by exhaustive search.

How to read this manual. Chapters 1 to 3 set out the rules and how the pieces behave. Chapters 4 and 5 form the tactical core: combinations and the triumph. Chapters 6 to 8 deal with the game plan, from the opening to the endgame. Chapter 9 annotates a complete game, and chapter 10 collects the puzzles, sorted by theme and difficulty. Chapters 2 to 8 end with the key points to remember.

Chapter 1

The Game

Two players, White and Black, face each other on a board of 8 × 8 squares. Each side has 16 pieces: 8 circles, 4 triangles and 4 squares, carrying the numbers 2, 3, 4 and 6. White moves first.

The Equipment

Shape	Pieces per side	Movement
Circle	8: two 2s, two 3s, two 4s, two 6s	1 square
Triangle	4: one 2, one 3, one 4, one 6	1 or 2 squares
Square	4: one 2, one 3, one 4, one 6	1, 2 or 3 squares

All pieces move in a straight line, in any of the eight directions, without jumping. Each number therefore exists in four copies: two circles, one triangle, one square.

The Starting Position

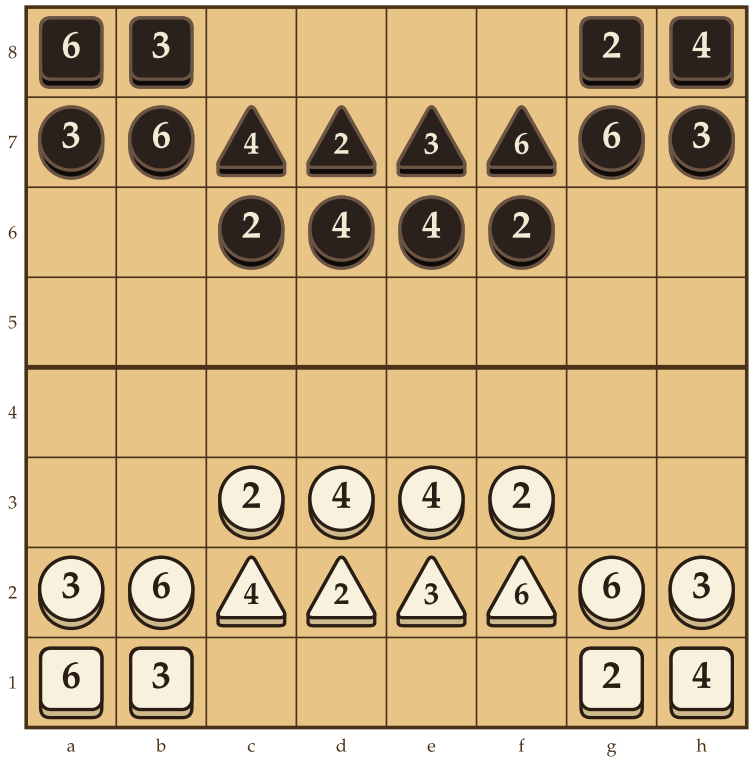


Diagram 1 — Starting position. Squares in the corners, triangles in the center of the second rank, circles 2 and 4 out in front. The thick line marks the middle of the board.

White: squares 6 a1, 3 b1, 2 g1, 4 h1; circles 3 a2, 6 b2, 6 g2, 3 h2; triangles 4 c2, 2 d2, 3 e2, 6 f2; circles 2 c3, 4 d3, 4 e3, 2 f3. Black is set up as a mirror image.

The Goal: the Triumph

A player wins when, after their move, **three of their pieces touch in a line** (rank, file or diagonal) **in the opponent's half of the board**, and their numbers form a progression, in either direction:

- 2-3-4 and 2-4-6 (arithmetic progressions: you always add the same amount);
- 2-3-6 and 3-4-6 (harmonic progressions).

The shape of the pieces does not matter: a circle 2, a square 3 and a triangle 4 form a triumph. The middle number is always a 3 or a 4.

Other Ways the Game Ends

- **Exhaustion.** A player who has only two different numbers left on the board can no longer form a progression: they lose.
- **Being blocked.** A player who has no legal move loses. This is very rare.
- **Draw.** After 60 consecutive plies (half-moves) without any capture, the game is a draw.

Captures

You capture an enemy piece in three ways, detailed in chapter 3:

- **Encounter** (equality): land on an enemy piece with the same number but a different shape.
- **Assault** (multiplication or division): land on an enemy piece worth double or half your own after moving exactly 2 squares, or triple or a third after moving 3.
- **Ambush** (addition or subtraction): after the move, the piece just moved and another friendly piece frame an enemy piece on a line; if the sum or difference of their numbers equals its number, it is captured.

Capturing is never compulsory. A single move can make several captures.

Notation

Squares are named from White's point of view: files a to h, ranks 1 to 8. White's opponent's half, where White must triumph, is made up of ranks 5 to 8. Black's is made up of ranks 1 to 4.

Written	Meaning
C3 a2-b3	circle 3 moves from a2 to b3
S2 g1×g4	square 2 captures the piece on g4 (Encounter or Assault)
C4 d3-d4 (×e5)	the move captures the piece on e5 by Ambush
#	triumph or exhaustion: the game is won
! / ?	good move / mistake

The letters stand for the shapes: C for circle, T for triangle, S for square. "S2" reads "square 2".

Chapter 2

The Pieces

A piece's worth comes from three things: its shape, which sets its range; its number, which sets what it captures and what captures it; and the scarcity of that number in its own army. Squares are the strongest pieces, circle 3 the weakest, and the last piece of a number can be worth the game.

Range

Shape	Squares reached from the center, empty board	Moves to enter the opponent's half
Circle	8	2 (circles 2 and 4) or 3 (circles 3 and 6)
Triangle	16	2
Square	24	2

A circle needs five to six moves to reach the back of the opponent's half; a triangle or a square needs three. Range also decides captures: only triangles and squares can make an Assault, and only squares can use the three-square "triple or a third" scale.

What Each Number Captures and Suffers

Each piece has only two kinds of enemies by Encounter: the two other shapes of its number. Beyond that, the numbers are not equal.

- **The 2 is the most active number.** By Assault, it captures 4s and 6s. It can play four different pincer roles in an Ambush.
- **The 2 and the 6 are the most exposed**: five types of pieces can capture them by moving.

- **The 3 is the least exposed**: only one pair of pincers (3 and 6) can capture it by Ambush.

Capture Balance

In the strong engine's 102 games, here is how many captures each piece makes and suffers, per 100 plies spent on the board.

Piece	Captures made	Times captured	Ratio
Square 2	1.25	0.62	2.0
Square 4	1.00	0.49	2.0
Square 3	1.04	0.54	1.9
Square 6	1.21	0.65	1.9
Triangle 4	1.13	0.63	1.8
Triangle 2	1.44	0.88	1.6
Triangle 3	0.94	0.64	1.5
Triangle 6	1.09	0.73	1.5
Circle 6	0.46	0.43	1.1
Circle 2	0.77	0.86	0.9
Circle 4	0.68	0.82	0.8
Circle 3	0.30	0.50	0.6

Squares capture twice as often as they are captured, triangles one and a half times as often, circles barely as often. Circles 2 and 4 make most of their captures by Ambush: they are pincers.

The Scale of Values

The unit is the loss of a circle 3 early in the game. The values come from a regression over 575 games: what one piece fewer costs in winning chances. They give an order of magnitude, not an exact scale.

Shape	2	3	4	6
Circle	2.5	1	2.5	2
Triangle	4	3.5	4	3.5
Square	4.5	4	3.5	4.5

Remember three tiers: a circle is worth 1 to 2.5, a triangle about 4, a square about 4 to 4.5. Squares 2 and 6, which can Assault two different numbers, are the most valuable pieces.

Scarcity

Each number exists in four copies. The more of them are missing, the more the survivors count, because they are the ones keeping that number alive.

Pieces of this number left, before the loss	Extra cost, in circles 3
4	+0.7
3	+0.8
2	+1
1 (the number disappears)	+2.5

Losing the last copy of a number therefore costs about two and a half pieces more than the piece itself.

The Royal Piece

When a number has disappeared completely, only three numbers remain. The last piece of each of them then becomes **royal**: capturing it drops the side to two numbers, and so loses the game. It plays the role of the king in chess.

Two royal pieces at once are almost impossible to defend: in the endgames tested, the side that had two scored only 5%. A side reduced to three numbers also has only one progression left with which to triumph.

Roles

Role	Pieces	Use
Strikers	T2, S2, S6, T4, T6	make the most captures; lead the attack
Anchors	S3, S4	rarely captured, active; hold the key squares
Support	C2, C4	Ambush pincers; highly exposed
Passive	C3, C6	rarely captured, not very active; wake them up early

Key points

A square is worth about four circles 3, a triangle a little less. Watch scarcity: the last piece of a number is worth far more than its own value, and once a number has disappeared, it is worth the game.

Chapter 3

The Three Captures

The three captures rest on three relations: equality (Encounter), product (Assault), sum or difference (Ambush). In strong games there are about 16 captures per game: 46% Encounters, 31% Ambushes, 23% Assaults.

The Encounter

A piece captures by landing on an enemy piece **with the same number and a different shape**, within range, along a clear path. Two identical pieces (two circles 4, for example) can never capture each other.

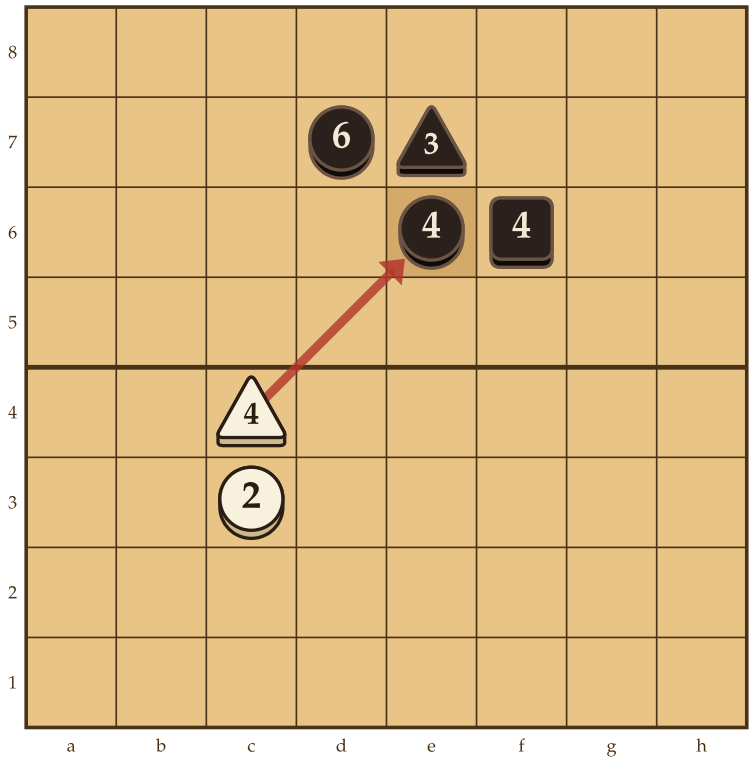


Diagram 2 — T4 c4×e6. The triangle 4 travels two squares diagonally and captures the circle 4. But the black square 4 immediately recaptures by Encounter: S4 f6×e6. White has traded a triangle for a circle, a bad deal.

The Encounter is the most frequent capture and the easiest to see. Each piece has exactly two kinds of enemies by Encounter: the two other shapes of its number. It is also the main weapon of **defense**: a piece protected by a friendly piece with the same number and a different shape can only be captured at the cost of a recapture.

The Assault

A piece with number v captures by landing on an enemy piece located **exactly k squares away** (k = 2 or 3, within its range, clear path) if the target is worth v × k or v ÷ k. Circles, which move only one square, can never make an Assault.

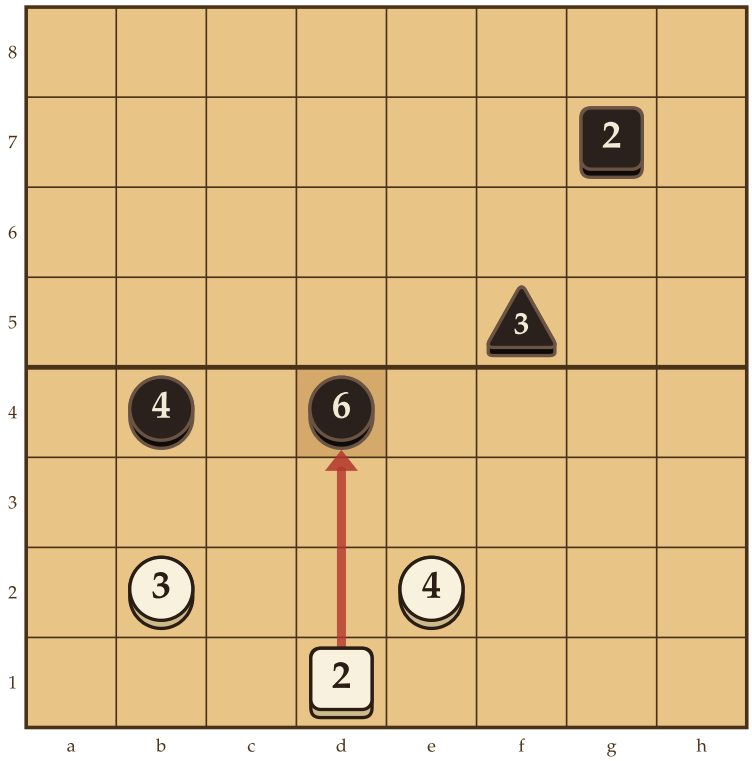


Diagram 3 — S2 d1×d4. The square 2 travels three squares: 2 × 3 = 6, so it captures the circle 6.

Attacker	Captures at 2 squares	Captures at 3 squares (square only)
2	a 4	a 6
3	a 6	—
4	a 2	—
6	a 3	a 2

A triangle threatens only one number by Assault. Squares 2 and 6 threaten two: that is what makes them the most valuable pieces. The distance is fixed: a triangle 2 captures a 4 two squares away, never one. You parry an Assault by stepping to the wrong distance or by blocking the line with an interposed piece.

The Ambush

After a move, if the piece just moved touches an enemy piece and a friendly piece stands **right behind it**, on the same line, the enemy piece is captured when its number is the **sum or difference** of the two pincers. Shape does not matter.

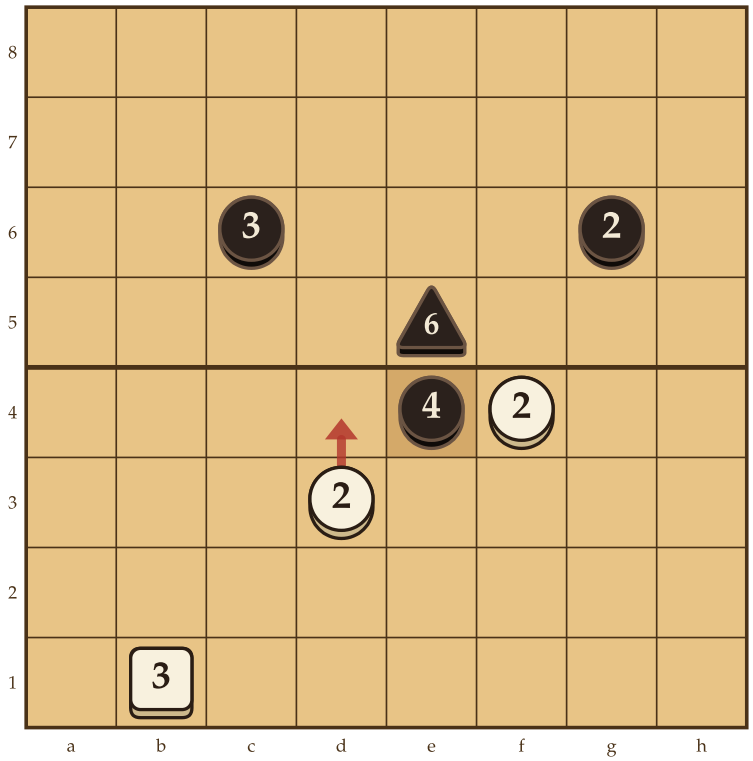


Diagram 4 — C2 d3-d4 (×e4). The circle 2 moves up against the black circle 4; the white circle 2 on f4 closes the pincer: 2 + 2 = 4, and the circle 4 is captured.

Target	Possible pincers
2	4 and 2 (4 − 2), 6 and 4 (6 − 4)
3	6 and 3 (6 − 3)
4	2 and 2 (2 + 2), 6 and 2 (6 − 2)
6	2 and 4 (2 + 4), 3 and 3 (3 + 3)

Three rule points matter in practice:

- **Only the piece just moved triggers the Ambush.** A piece that moves in between two enemies is in no danger.

- **The edge protects.** A piece against the edge cannot be caught in a pincer attack in directions that leave the board. A corner is vulnerable in only three directions out of eight.
- **A friendly piece behind you also protects**, because the square where the second pincer would have to stand is occupied.

Exchanges

A capture followed by a recapture is an exchange. You judge it with the scale from chapter 2. In diagram 2, White loses a triangle (4) for a circle (2.5). Before capturing, count who can recapture on the arrival square: the two other shapes of the same number by Encounter, Assaults from two or three squares, and nearby pincers.

> Key points
>
> The Encounter attacks and defends, the Assault strikes from afar at a fixed distance, and the Ambush is prepared by placing the second pincer in advance. Before every capture, ask yourself who recaptures.

Chapter 4

Tactics

A combination wins material or the game through a forced sequence of at least two moves. Combinations exist in about 6% of positions, as often as in chess. Among the 142 verified material-winning puzzles, the double attack dominates (69), ahead of the triumph threat (35), the quiet move that prepares a capture (26), the sacrifice (22) and the discovered attack (17).

Seeing Threats

Each piece threatens according to three different rules, and threats hide easily. Before every move, go through three questions, for yourself and for your opponent.

1. **Who can land on whom?** For each piece, the two other shapes of its number (Encounter), and the numbers it captures at exactly 2 or 3 squares (Assault).
2. **Which pincer can close?** An enemy piece touched on one side by an empty square and on the other by one of your pieces is in danger if a piece with the right number can come to the empty square.
3. **Which triumph line has two pieces out of three?** Such a line, if the third piece can arrive, is a threat to win at once.

The Double Attack

A single move creates two threats; the opponent can parry only one.

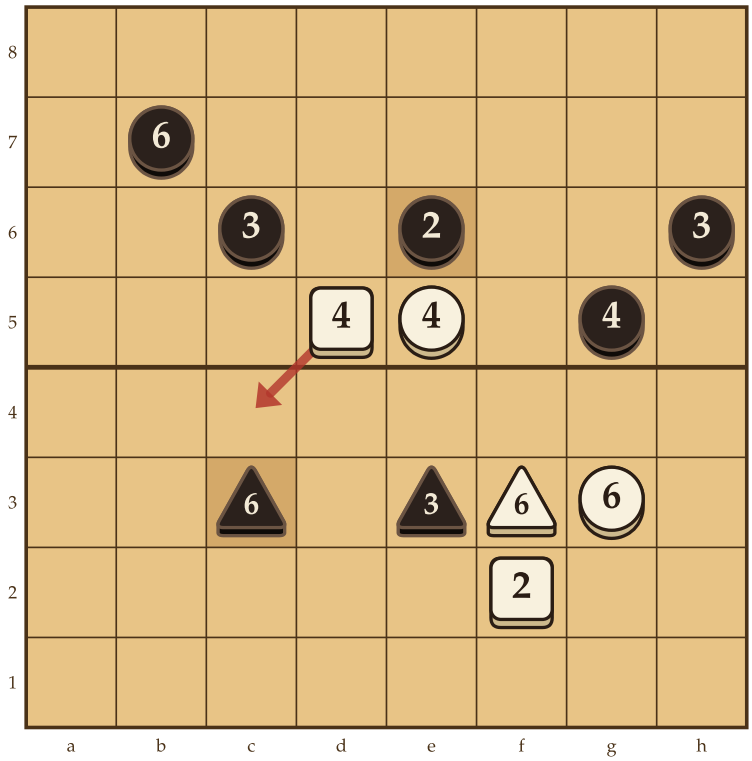


Diagram 5 — White to move. White: S4 d5, C4 e5, T6 f3, C6 g3, S2 f2. Black: C6 b7, C3 c6, C2 e6, C3 h6, C4 g5, T6 c3, T3 e3.

The white square 4 is threatened: T6 c3-c4 would capture it by Ambush ($6 - 2 = 4$, with the circle 2 on e6 behind it). **S4 d5-c4!** saves the square and creates two threats:

- S4 c4×e6, an Assault from two squares ($4 \div 2 = 2$);
- S2 f2-c2, which catches the triangle 6 in a pincer attack ($2 + 4 = 6$, with the square 4 behind it).

If the circle 2 runs away, the triangle falls. If the triangle runs to d3, S2 f2-e2 catches it again in a pincer attack against the same square 4: $2 + 4 = 6$. White wins a piece.

The Triumph Threat, the "Check" of Rithmomachia

A triumph threat forces the opponent to respond, like check in chess. You use it to win material: the piece that comes to block the line is often lost.

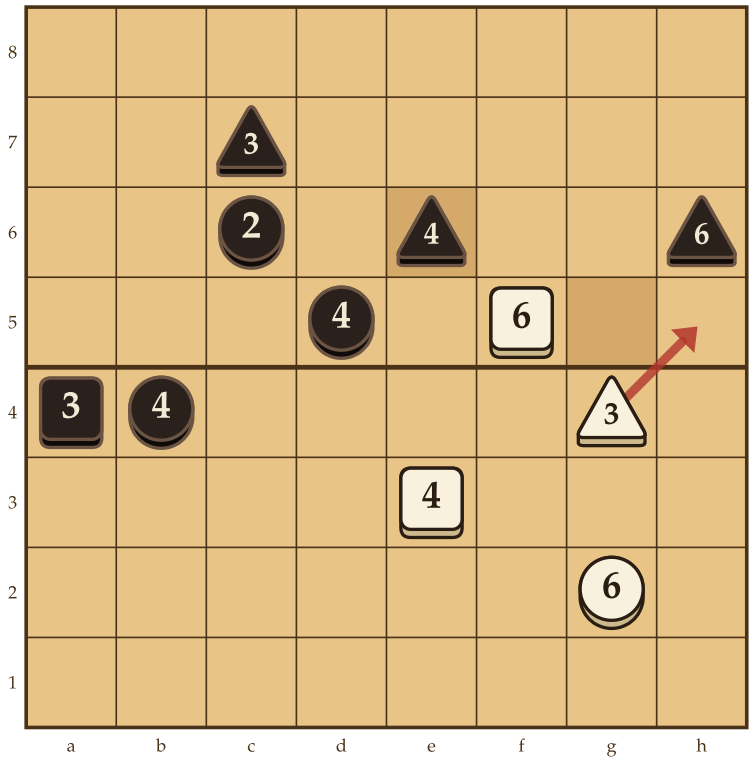


Diagram 6 — White to move. White: T3 g4, S6 f5, S4 e3, C6 g2. Black: T3 c7, C2 c6, T4 e6, T6 h6, C4 d5, S3 a4, C4 b4.

T3 g4-h5! threatens S4 e3-g5, which would line up S6 f5, S4 g5 and T3 h5 on the fifth rank: 6-4-3, triumph. Black must interpose the triangle 6, on g5 or on f4. Either way, the square 6 captures it by Encounter (S6 f5×g5 or S6 f5×f4). Any other reply loses at once. And the threat S4 e3×e6 against the triangle 4 is still held in reserve.

The Discovered Attack

A piece leaves a line and opens the way for another. Since Assaults and Encounters reach far, a single moved piece can unleash several threats at once.

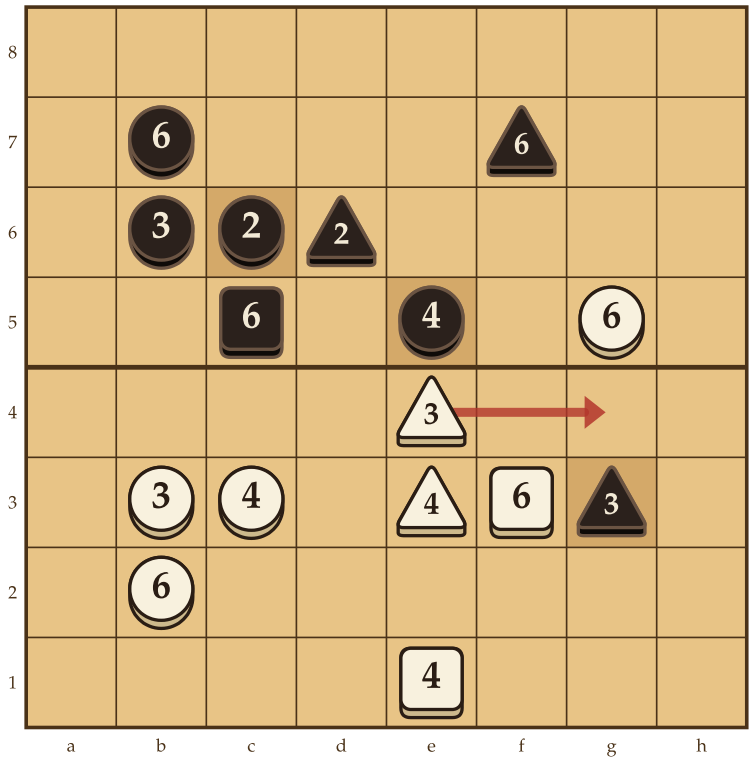


Diagram 7 — White to move. White: T3 e4, C6 g5, C3 b3, C4 c3, T4 e3, S6 f3, C6 b2, S4 e1. Black: C6 b7, T6 f7, C3 b6, C2 c6, T2 d6, S6 c5, C4 e5, T3 g3.

Black threatens T3 g3×g5 ($3 \times 2 = 6$). **T3 e4-g4!** closes the g-file and protects the circle 6. By leaving e4, the triangle uncovers two lines:

- T4 e3×e5 (Encounter on the circle 4);
- S6 f3×c6 (Assault from three squares, $6 \div 3 = 2$).

On top of that, the black triangle 3 is caught in a net: S6 f3-g2 or T3 g4-h3 would catch it in a pincer attack ($6 - 3 = 3$). If it runs to h3, S6 f3×h3 captures it by Assault ($6 \div 2 = 3$). Black cannot save everything.

Deflection and Sacrifice

You offer a piece to drag a defender away. Here the defender is blocking a line: if it moves, the line opens.

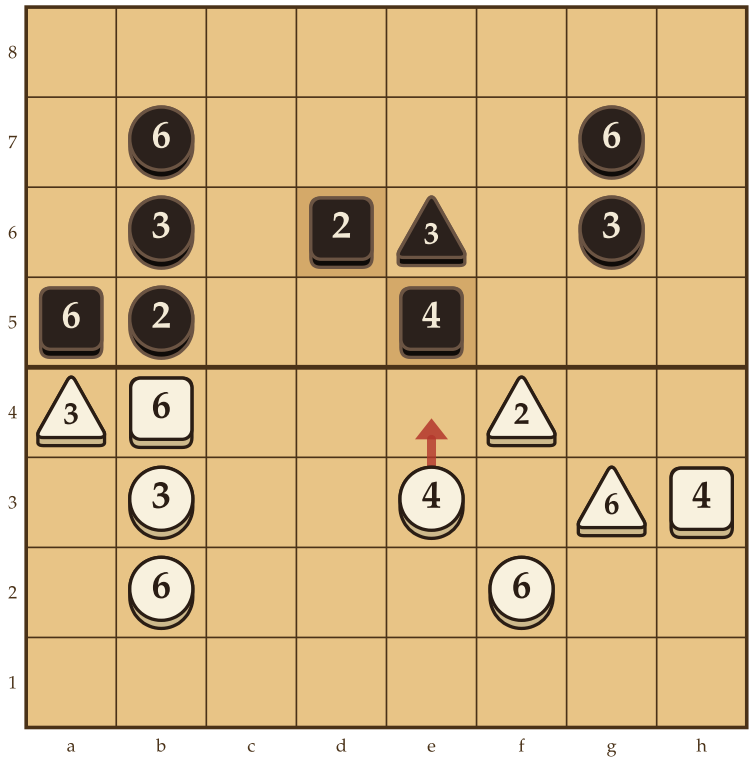


Diagram 8 — White to move. White: T3 a4, S6 b4, T2 f4, C3 b3, C4 e3, T6 g3, S4 h3, C6 b2, C6 f2. Black: C6 b7, C6 g7, C3 b6, S2 d6, T3 e6, C3 g6, S6 a5, C2 b5, S4 e5.

The black square 4 on e5 threatens S4×e3. **C4 e3-e4!** attacks it by Encounter. The square is blocking the f4-e5-d6 diagonal: if it captures (S4 e5×e4) or steps aside, T2 f4×d6 captures the square 2 by Encounter. If it is the square 2 that runs, C4×e5 captures the square 4. At worst, White gives a circle 4 (2.5) for a square 2 (4.5).

The Pincer

An Ambush is prepared: first you place the second pincer, then you bring in the first. Circles 2 and 4 are the natural pincers.

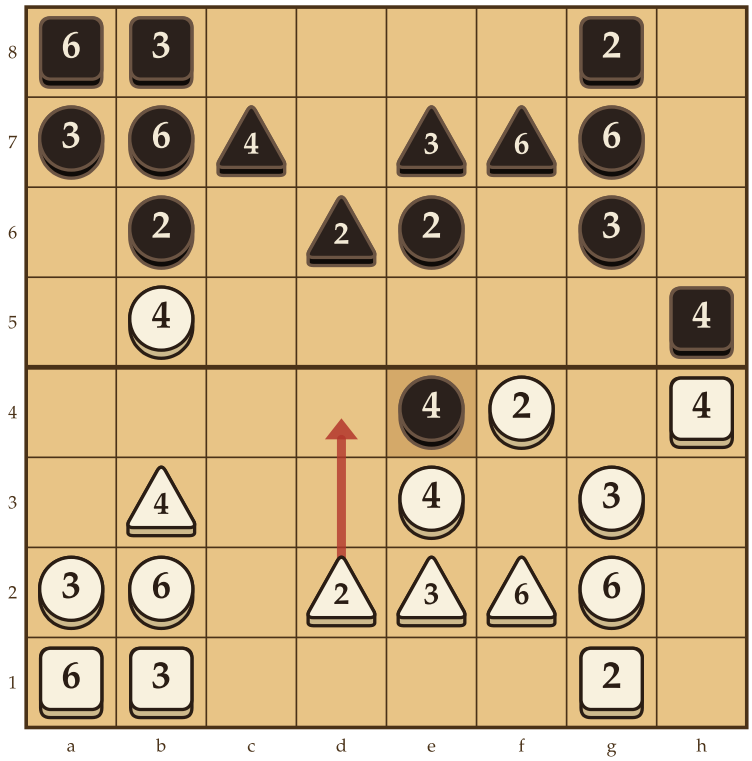


Diagram 9 — White to move, early in the game. The black circle 4 has advanced to e4, in contact with the white circle 2 on f4.

T2 d2-d4! closes the pincer: the circle 4 is captured ($2 + 2 = 4$). The triangle is already eyeing the circle 2 on b6. If Black replies T2 d6×f4, the square 4 recaptures by Assault (S4 h4×f4, $4 \div 2 = 2$).

A circle pushed into contact with an enemy piece, with an empty square behind it, is the typical target of a pincer attack. Before advancing a circle, look at which pincers could close on it.

Key points

Look for the move that creates two threats. The triumph threat also serves to win material. A piece blocking a line is an overloaded defender: attack it. And every piece you move can open a line, for you or against you.

Chapter 5

The Triumph

More than eight games in ten end in a triumph. It most often forms just past the center line, and it is prepared two moves ahead by a double threat. By the loser's last move, there was already no defense in 70% of cases.

The Role of the Numbers

Progression	Middle	Ends
2-3-4	3	2 and 4
2-3-6	3	2 and 6
2-4-6	4	2 and 6
3-4-6	4	3 and 6

The middle is always a 3 or a 4; the ends are often a 2 and a 6. Each number belongs to three of the four progressions. If a number disappears, only one progression remains possible.

The Geometry of the Zone

The opponent's half contains 64 lines of three squares. Here is how many pass through each square, seen from White's side.

Rank	a	b	c	d	e	f	g	h
8	3	4	6	6	6	6	4	3
7	4	7	9	9	9	9	7	4
6	4	7	9	9	9	9	7	4
5	3	4	6	6	6	6	4	3

The central squares of ranks 6 and 7, that is, the opponent's starting ranks, are the richest. A piece placed there can be the middle of four lines at once: rank, file and two diagonals. On ranks 5 and 8, a piece is the middle of only one line, the rank. So ranks 6 and 7 are where you should place your 3s and 4s.

Where Triumphs Form

In the strong engine's games:

Criterion	Distribution
Rank in the opponent's half	5: 58% · 6: 22% · 7: 18% · 8: 2%
Orientation	rank 57% · diagonal 28% · file 16%
Progression	2-3-4: 30% · 3-4-6: 28% · 2-4-6: 27% · 2-3-6: 16%
Files	c to f: 61% · b and g: 24% · a and h: 14%
Finishing piece	square 40% · triangle 36% · circle 24%
Last move	one square: 65% · with a capture: 22%

The typical triumph is therefore a row of three pieces set just past the center line, in the middle of the board, completed by a square moving one square.

Two-Piece Formations

Two well-placed pieces are a threat in the making. Not all are equal: what counts is the number of squares where the third piece can finish the line.

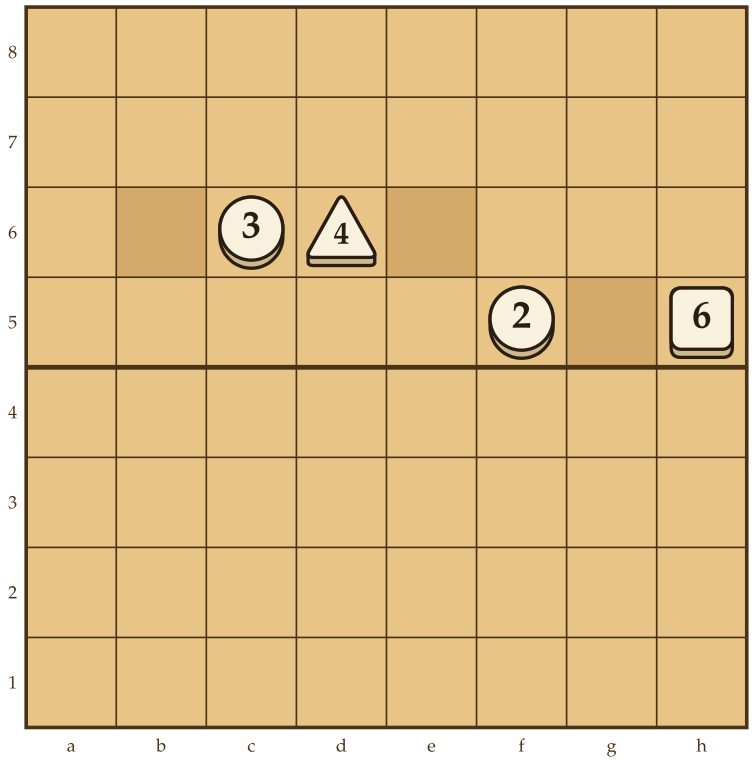


Diagram 10 — The 3-4 duo (c6, d6) can be completed on both sides: a 2 on b6 or a 6 on e6. The 2 _ 6 frame (f5, h5) accepts a 3 or a 4 in the middle, on g5.

Formation	Completions	Squares to hold on defense
3-4 duo side by side	a 2 on the 3's side, a 6 on the 4's side	two
2 _ 6 frame	a 3 or a 4 in the middle	one, but two possible numbers
2-3 or 4-6 duo	two numbers, on the same side	one
2-4 or 3-6 duo, 2 _ 4 and 3 _ 6 frames	a single number	one

The 3-4 duo is the strongest formation: it threatens two squares at once.

The Double Threat

A single threat is parried by blocking the completion square. To win, you must create two threats on two different squares in a single move.

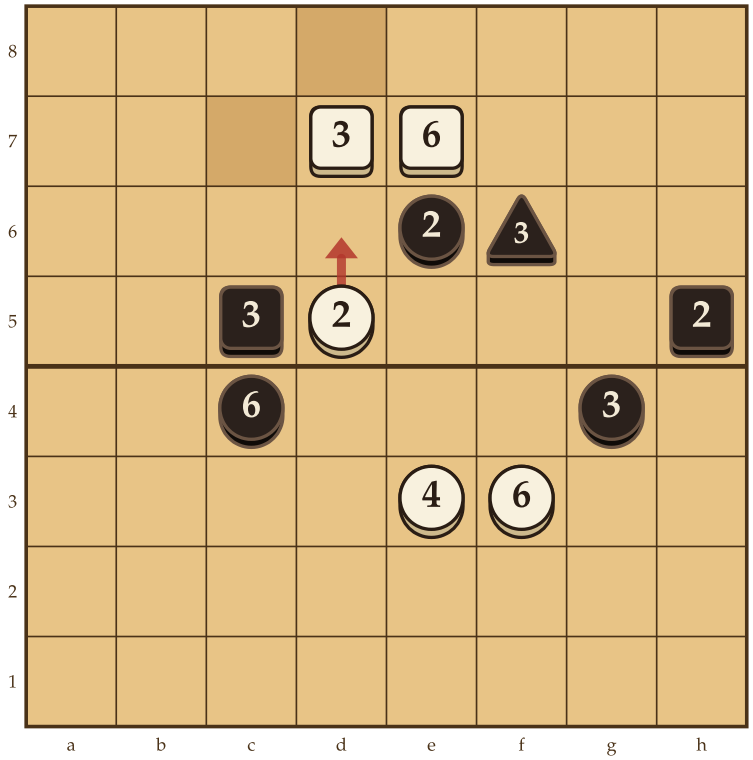


Diagram 11 — White to move. White: S3 d7, S6 e7, C2 d5, C4 e3, C6 f3. Black: C2 e6, T3 f6, S3 c5, S2 h5, C6 c4, C3 g4.

C2 d5-d6! The square 3 on d7 is in the middle of two lines. Two triumphs are threatened:

- C2 d6-c7, forming c7-d7-e7 = 2-3-6 on the seventh rank;
- S6 e7-d8, forming d8-d7-d6 = 6-3-2 on the d-file.

Black can block only one square, and can capture neither the circle 2 nor the square 3. The triumph follows next move. Note the d7 square: it is one of the four-line squares of rank 7.

Defending

- **Count the two-out-of-three lines in your half**, especially on your rank 4: that is where the opponent triumphs most often.
- **Block the completion square** with a piece that cannot be captured there. Otherwise, as in diagram 6, the interposed piece falls.
- **Prefer to capture the middle piece**, a 3 or a 4: it often belongs to several lines.
- **Think one move ahead.** A double threat can no longer be parried. You must prevent the position that allows it: deny the key square or capture one of the two pieces of the duo.
- **Hold your ranks 3 and 4** (chapter 6): a square occupied by your piece is a square the opponent cannot use.

> Key points
>
> Aim for the rank just past the center line, keep your 3s and 4s for the middle of lines, build a 3-4 duo and look for the move that threatens two squares. On defense, see the double threat one move before it arrives.

Chapter 6

Strategy

Modern Rithmomachia is a positional game as much as a material one: an engine that counts only pieces loses 40 games out of 40 against the same engine that understands position. Four principles held up against the strong engine: advance into the opponent's half, hold your front line, keep your four numbers, and prepare triumph lines.

What Wins, in Order of Importance

Each principle was tested by having the engine that knows it play against the same engine that ignores it (40 games per test).

Principle	Score of the side applying it	Verdict
Advance into the opponent's half	92%	decisive
Hold your front line (your ranks 3 and 4)	78%	very strong
Keep your four numbers	76%	very strong
Prepare lines with two pieces out of three	68%	strong
Group your pieces, avoid the edges	50%	no effect

These scores slightly overstate each principle: the engine deprived of it had been tuned to use it. The order, however, is reliable.

Advancing

You can only win in the opponent's half. A piece settled there threatens triumphs, forces the opponent to respond and blocks their starting ranks. In the endgame, presence in the opponent's half weighs as much as material. It counts even more

when pieces advance in pairs, because two neighboring pieces already make up two-thirds of a triumph.

The Front Line

Your ranks 3 and 4 are both your rampart and your springboard. The opponent forms most of their triumphs on your rank 4: every square you occupy there is denied to them. And it is from these ranks that your pieces enter the opponent's half in one move.

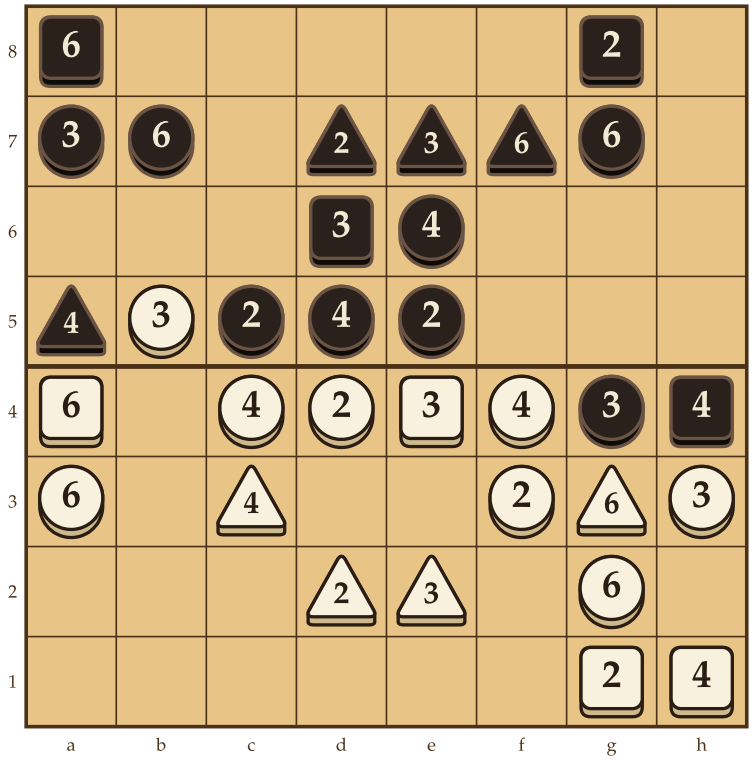


Diagram 12 — Strong engine game, after 12 moves; White to move, and White will win. White holds rank 4 from a to f. White's squares 6 and 3 are out (a4, e4); the triangles 2 and 3 stay in reserve on d2 and e2. The circle 3 on b5 is already in the opponent's half.

The front line matters in every phase. In the opening, at the strong level, it weighs as much as material.

Squares Out, Triangles in Reserve

In strong games, two factors stand out in the opening and middlegame:

- **developing your squares early pays off.** They have the longest range, capture twice as often as they are captured, and finish 40% of triumphs;
- **developing your triangles early costs you.** This factor weighs almost as much as the front line, but negatively. A triangle advanced too early becomes a target; kept on the second rank, it protects and bides its time.

In the endgame, the trend reverses: every piece must join the attack.

Keeping Your Four Numbers

As long as a side has its four numbers, none of its pieces is vital and it has every progression available. As soon as a number disappears, one or two pieces become royal and only one progression remains. Before an exchange, count how many copies of each number you will have left. With equal pieces, four numbers against three scores 75% in the endgame (chapter 8).

Preparing Lines

A line with two pieces out of three in the opponent's half, whose third piece can arrive, is a permanent threat. It ties down the opponent's defenders and prepares the double threat from chapter 5. In the middlegame, it weighs as much as the front line.

What Matters Little

Grouping your pieces in pairs, avoiding isolated pieces or shunning the edges did not win games at the strong level. The a- and h-files even serve as corridors for the squares, as in the opening (chapter 7). An edge also protects against Ambush. Do not give up one of the four principles above for the sake of shape.

> Key points
>
> Advance, hold your ranks 3 and 4, keep your four numbers and prepare lines with two pieces out of three. Bring your squares out; save your triangles for later.

Chapter 7

Openings

The opening has one goal: to wake up the sleeping pieces (the wing circles 3 and 6, the corner squares) without stripping ranks 3 and 4. The strong engine almost always does this the same way, with the corner plan. No rigid plan does better than flexible play.

The Plans Tested

Each plan imposes one side's first four moves, then the engine plays freely. Score of the side following the plan (40 games at the strong level, 100 at the medium level).

Plan	Moves (seen from White)	Medium level	Strong level
Corners	C3 a2-b3, C3 h2-g3, S6 a1-a4, S4 h1-h4	46%	49% vs. free play; 57% vs. "Wings"
Wings	C6 b2-b3, C6 g2-g3, S6 a1-b2, S4 h1-g2	57%	34% vs. free play
Center	C4 d3-d4, C4 e3-e4, T2 d2-d3, T3 e2-e3	50%	55% vs. "Wings"
Wall	C2 c3-c4, C4 d3-d4, C4 e3-e4, C2 f3-f4	45%	40% vs. free play

The margins are wide (about ±15 points at the strong level), but two conclusions are clear. The "Wings" plan, which looked best at the medium level, is bad at the strong level. And the wall of circles is bad at every level.

The Corner Plan

Left to itself, the strong engine almost always plays the same moves, for both sides:

1. **C3 a2-b3 and C3 h2-g3.** The circles 3 are the most passive pieces in the game. By stepping up diagonally, they join the front line and free the a- and h-files.
2. **S6 a1-a4 and S4 h1-h4.** The corner squares come out along the edge file, where nothing gets in their way, and post themselves on rank 4.

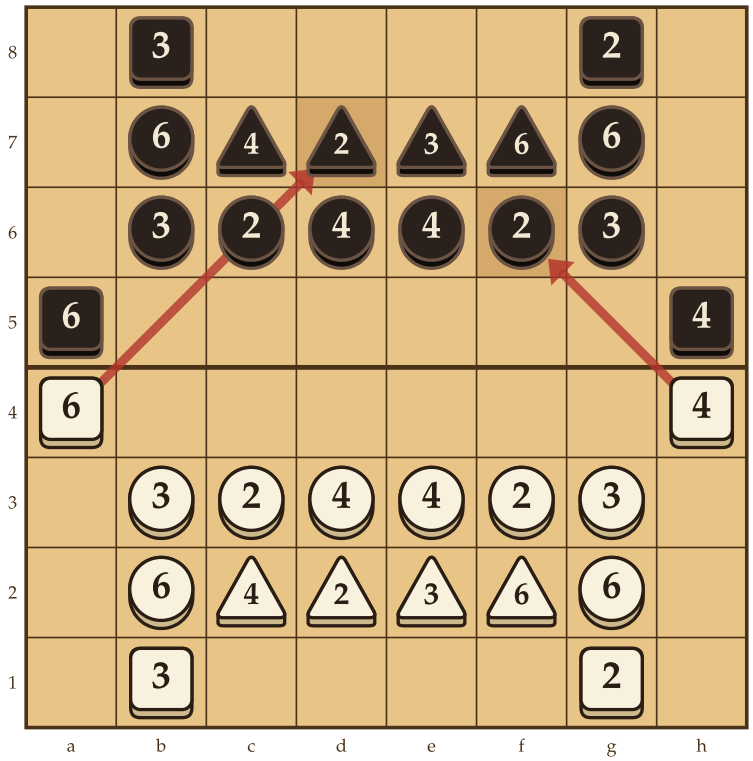


Diagram 13 — Both sides have played the corner plan. The white square 4 eyes the circle 2 on f6 for an Assault ($4 \div 2 = 2$), but the circle 4 on e6 defends it. The square 6 eyes the long diagonal toward d7: as soon as the circle 2 on c6 moves, the triangle 2 is threatened by an Assault from three squares ($6 \div 3 = 2$). Black exerts the same pressure in mirror image.

The two developed squares immediately bear down on the opponent's circles 2, which are the opponent's pincers. The square 6 then often sets off on a raid toward d7. Meanwhile, the triangles stay on the second rank, in line with chapter 6.

Plans to Avoid

- **The wall of circles** (c4, d4, e4, f4) strips rank 3 and offers four targets in contact with the opponent. These are advanced circles, the typical prey of the pincer attack.

- **The forced wings plan** brings the squares out on the diagonal, behind their own circles. They stay shut in there longer and leave the circles 3 asleep.
- **Triangles first**: developing them early costs you at the strong level (chapter 6).

The Principles

- First wake up the corner circles 3, then bring out the corner squares.
- Keep your triangles in reserve and your ranks 3 and 4 manned.
- Do not push the central circles into contact without support.
- Stay flexible: move order matters less than answering the opponent's threats. Imposed blindly, even the corner plan does no better than free play.

> Key points
>
> C3 a2-b3, C3 h2-g3, then the corner squares along the a- and h-files. Triangles at home, ranks 3 and 4 held, no wall of circles.

Chapter 8

Endgames

In the endgame, the variety of numbers counts more than the number of pieces, squares dominate, and a royal piece becomes the center of the game, like the king in chess. Even with few pieces, most games still end in a triumph: attacking the opponent's half remains the goal.

What Advantages Are Worth

Each situation was replayed from 60 pairs of randomly drawn positions, each side in its own half, with colors reversed (120 games at the strong level). Score of side A.

Situation	Side A	Side B	A's score
Control, even	C2 C6 T3 T4 S2	same	51%
4 numbers vs. 3, equal pieces	C2 C6 T3 T4 S2	C2 C3 T3 T4 S2	**75%**
4 numbers vs. 3, one piece fewer	C2 C6 T3 T4 S2	C2 C3 T3 T4 S2 C4	50%
One extra circle	C2 C6 T3 T4 S2 C4	C2 C6 T3 T4 S2	57%
Squares vs. triangles	S2 S3 S4 S6 C2	T2 T3 T4 T6 C2	**94%**
Two royal pieces (no 6)	C2 T3 S4 C4 T4	C2 C6 T3 T4 S2	**5%**

Three lessons:

- **Four numbers are worth more than an extra circle.** With equal pieces, the side with four numbers scores 75%. It takes one piece fewer to bring it back to even.
- **Squares crush triangles.** In the endgame, range decides: a square reaches the triumph zone and comes back to defend faster.
- **Two royal pieces means a lost game.** Never let two of your numbers drop to a single piece when a third has already disappeared.

Hunting the Royal Piece

When the opponent has only three numbers left, the game changes character: each of their unique pieces is worth the game. As in chess, you then attack the "king." In strong games, the final capture of an exhaustion win is made by Encounter in 64% of cases, by Ambush in 29%, by Assault in 7%.

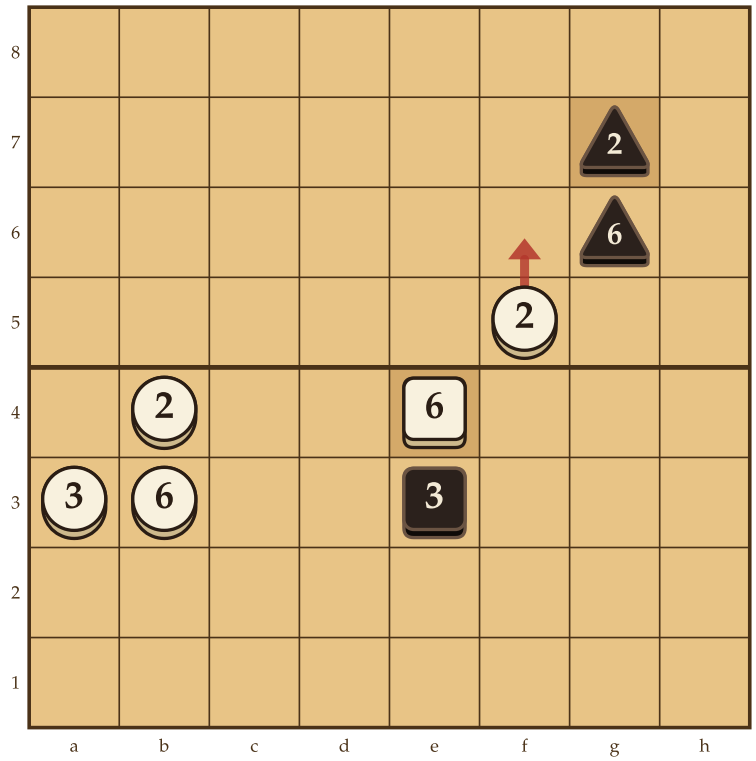


Diagram 14 — White to move. White: C2 f5, C2 b4, C3 a3, C6 b3, S6 e4. Black: T2 g7, T6 g6, S3 e3. Black has only three numbers left, each in a single copy: all three pieces are royal.

C2 f5-f6! attacks the triangle 2 by Encounter and, by leaving f5, uncovers the square 6's diagonal toward g6. Two royal pieces are threatened:

- if the triangle 2 runs away, S6 e4×g6 captures the triangle 6 and Black has only two numbers left;
- if Black captures the square (T6 g6×e4), C2 f6×g7 captures the triangle 2: exhaustion.

Defending a Royal Piece

- **Keep it away from the two other shapes of its number**, its only enemies by Encounter, and from Assault lines at two or three squares.
- **Put it against the edge or shelter it behind a friendly piece**: it then escapes Ambush in those directions.
- **Keep it away from discovered attacks**: do not leave it on a line where only a single enemy piece stands in between.
- **Prevention is better**: before exchanging, count your copies of each number.

A Piece Down

In the medium-level corpus, the side that is one piece down in the endgame still wins 36% of games; two pieces down, 24%; three pieces down, 12%. A material deficit is therefore not final, especially if you keep your four numbers and pieces in the opponent's half.

> Key points
>
> Keep your four numbers, prefer squares, never allow two royal pieces. Against three numbers, hunt the royal piece with the double attack and the discovered attack.

Chapter 9

Annotated Game

This game pits the strong engine against itself. White wins by triumph on move 31. It features the corner plan, two double captures and a finish by double threat. The first two moves for each side were drawn at random to vary the games; the annotations rely on the engine's five-ply analysis.

The Opening: the Corner Plan on Both Sides

> **1. C2 f3-e4 C4 e6-f5 2. T3 e2-e1 C4 d6-e6** (*randomly drawn moves*) **3. C3 a2-b3 C6 g7-h6 4. S6 a1-a4 C3 a7-b6 5. C6 g2-h3 S6 a8-a5**

Both sides wake up their wing circle 3 and bring their square 6 out along the a-file. The squares 6 face each other on a4 and a5: being identical, they cannot capture each other.

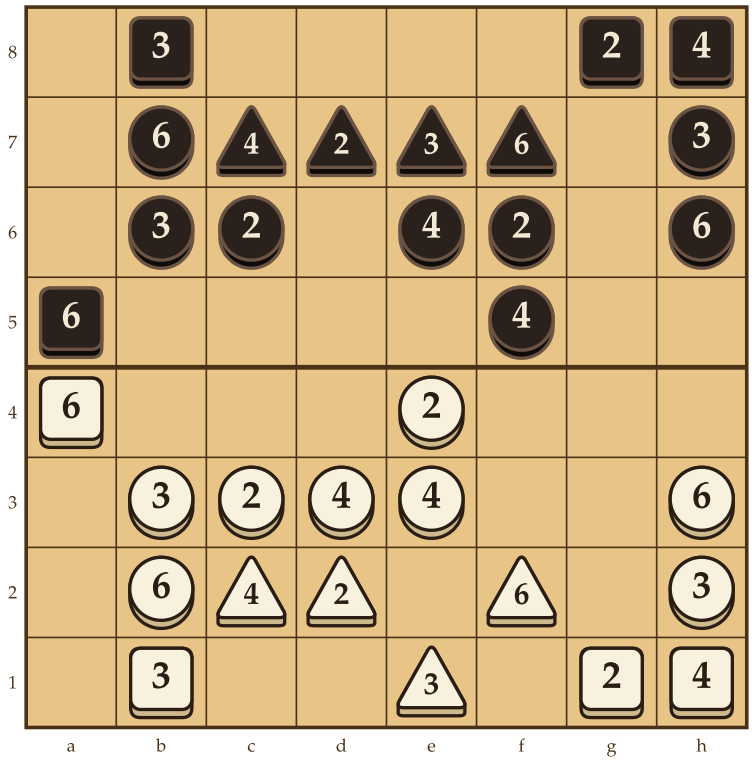


Diagram 15 — After 5... S6 a8-a5.

> **6. S6 a4-c4?! S2 g8-g5 7. T6 f2-h4 C6 h6-g7 8. S2 g1-g4 C4 e6-e5?!**

The square 6 leaves its file for c4; it will return on move 12. That is a lost tempo, and the analysis preferred developing the square 4. On the other side, 8... C4 e6-e5 pushes a circle into contact without support: this is the typical mistake from chapter 4, and this circle will fall.

> **9. C4 e3-f4 S2 g5-g6 10. T3 e1-e3 T3 e7-d6 11. C4 f4-g5 T6 f7-d5 12. S6 c4-a4 C6 b7-a6? 13. S2 g4-f3 S2 g6-e8 14. S4 h1-g1 T2 d7-e6 15. C3 b3-c4 C6 a6-b5 16. S6 a4-a3 C3 h7-h6 17. S4 g1-g3 S3 b8-a7 18. C2 c3-d4 S2 e8-h5 19. C2 d4-c5 C2 c6-b7**

White has installed a circle 4 on g5 and a circle 2 on c5, in the opponent's half, while keeping rank 3. Black, for its part, has pulled back its squares (e8, h5) and its circles (a6). According to the analysis, White's advantage is already clear.

The Middlegame: Two Double Captures

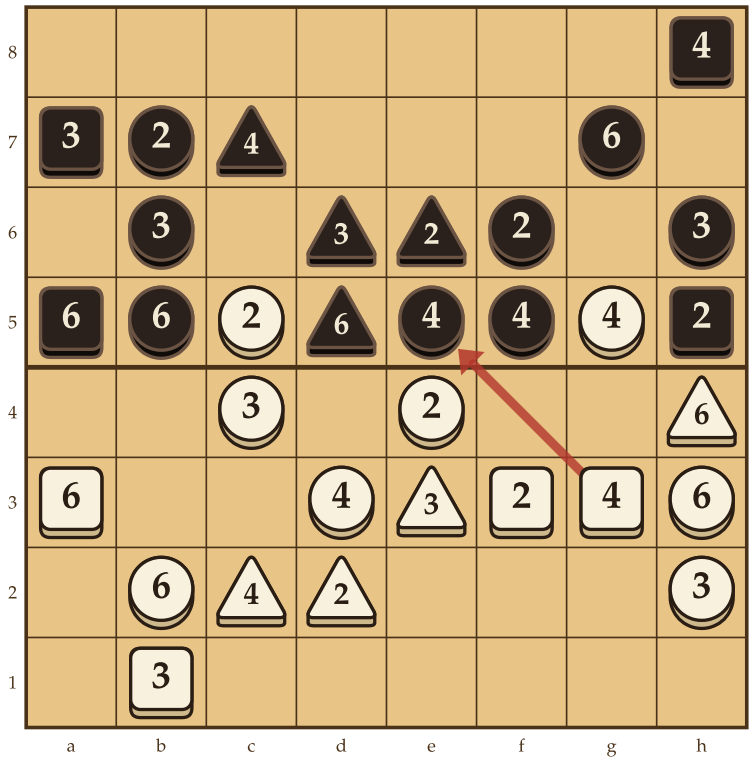


Diagram 16 — After 19... C2 c6-b7; White to move.

> **20. S4 g3×e5 (×d5)!**

The square 4 travels two squares and captures the circle 4 by Encounter. Arriving on e5, it touches the triangle 6 on d5, and the white circle 2 on c5 stands right behind it: 4 + 2 = 6, so the triangle falls too. The circle advanced on move 8 was indeed the target.

> **20... C4 f5×e5 21. C2 e4-d5 T2 e6-f5 (×g5) 22. S2 f3×f5 (×e5) C2 f6×f5**

Black recaptures the square, then catches the circle 4 on g5 in a pincer attack (2 + 2 = 4, with its square 2 on h5). The white square 2 answers with a second double capture: Encounter on the triangle 2 on f5, and Ambush on the circle 4 on e5 against the circle 2 on d5 (2 + 2 = 4). At the end of these exchanges, White has captured four pieces for three, and keeps a circle on d5, in the opponent's half.

> **23. S6 a3-b3 S6 a5-b4 24. C4 d3-d4 S3 a7-a4 25. T3 e3-e5 T4 c7-c6 26. T4 c2-e4 S2 h5-g5 27. T4 e4-f3 C2 f5-e4 (×d5)?**

The white triangle 3 settles on e5, next to the circle 2 on d5: a 2-3 duo on the fifth rank, waiting only for a 4 on f5. Black catches the circle 2 in a pincer attack, but this capture loses.

The Finish

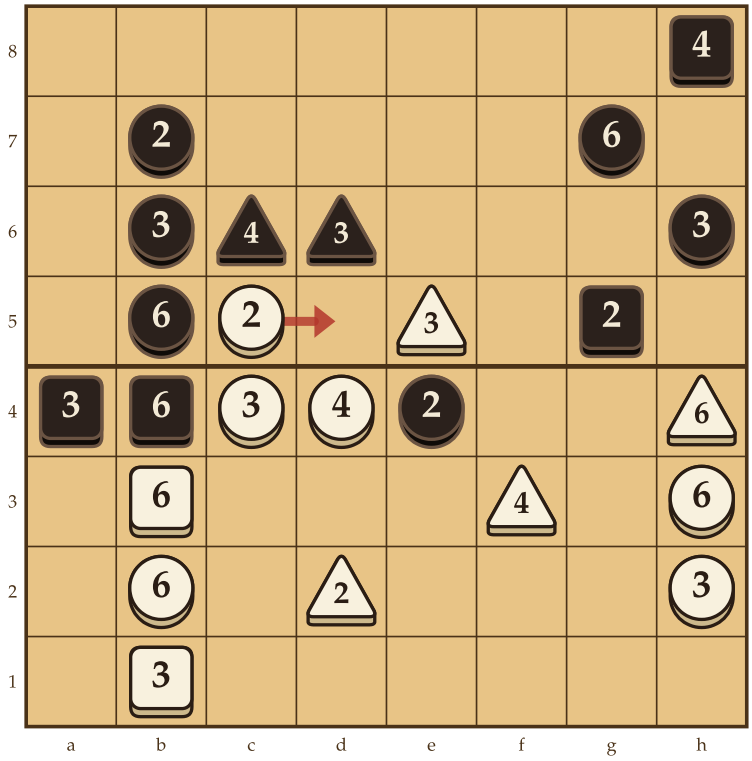


Diagram 17 — After 27... C2 f5-e4 (×d5); White to move.

28. C2 c5-d5 (×e4)!

The second circle 2 takes the place of the first. Arriving on d5, it catches the black circle 2 on e4 in a pincer attack against the triangle 4 on f3 ($4 - 2 = 2$). And it re-forms the d5-e5 duo: T4 f3-f5 threatens to complete 2-3-4. Black has no defense left.

28... S4 h8-g8 29. T6 h4-h5 C2 b7-a8 30. S3 b1-e4 C2 a8-b7 31. T4 f3-f5 (×g5) #

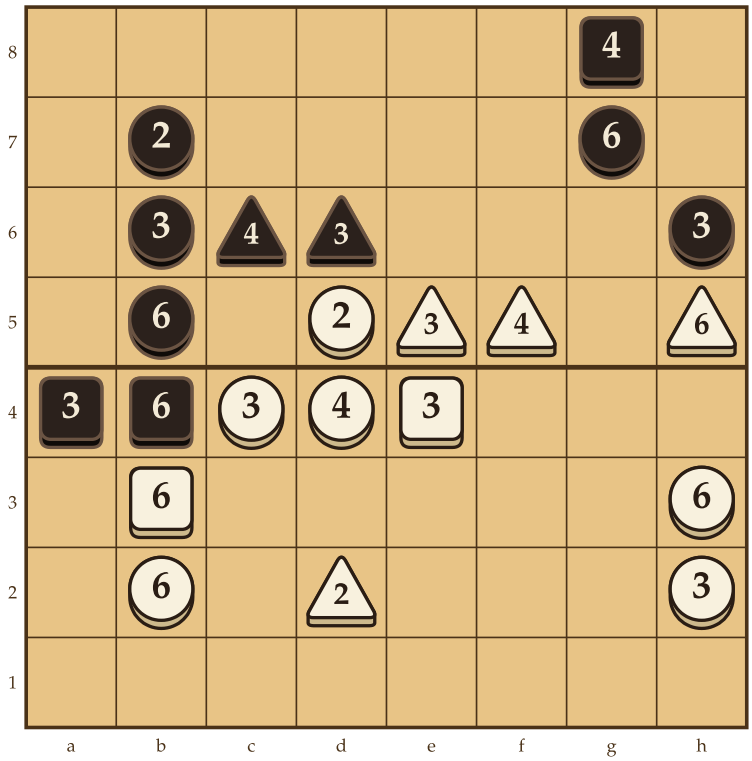


Diagram 18 — Final position. C2 d5, T3 e5, T4 f5: 2-3-4 on the fifth rank. On arrival, the triangle 4 also caught the square 2 on g5 in a pincer attack ($6 - 4 = 2$, against the triangle 6 on h5).

The Lessons

- **The corner plan** brought out the squares 6 on both sides by move 5.
- **The circle advanced without support** (8... C4 e6-e5) ended up taken in a double capture.
- **Presence in the opponent's half** was decisive: the circles 2 on c5 and d5, then the triangle on e5, formed the winning line on rank 5, where most triumphs form.
- **Double captures** come from a friendly piece already placed behind the target: they can be prepared.

Chapter 10

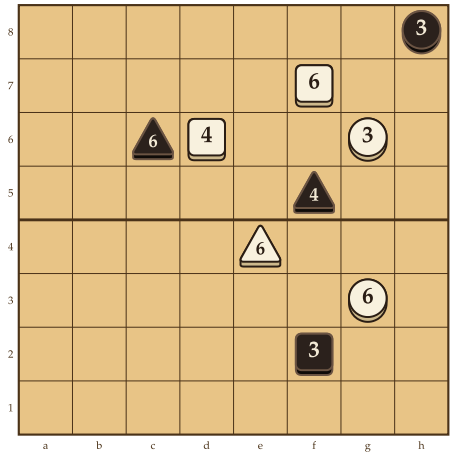
Puzzles

All these puzzles come from real engine games. Each has a single correct answer, verified by exhaustive search. In all of them, **White to play**. The solutions are at the end of the chapter.

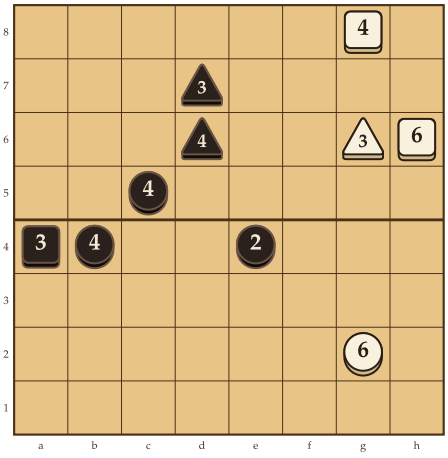
A. Win in two moves

White plays a move that no defense can parry, then wins on the next move, by triumph or by exhaustion. Look for the double threat, or the threat that nothing can block. Every defense is covered in the solutions.

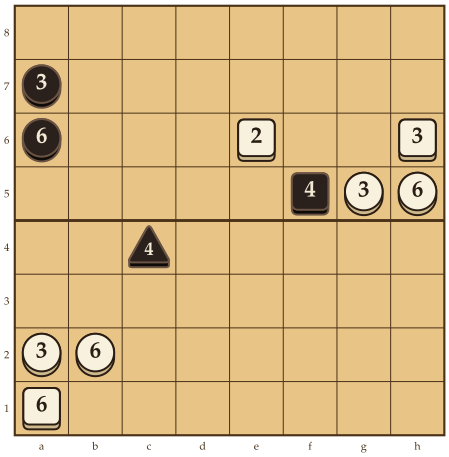
Puzzle 1



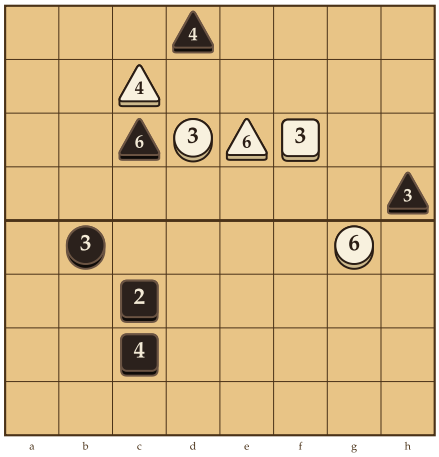
Puzzle 2



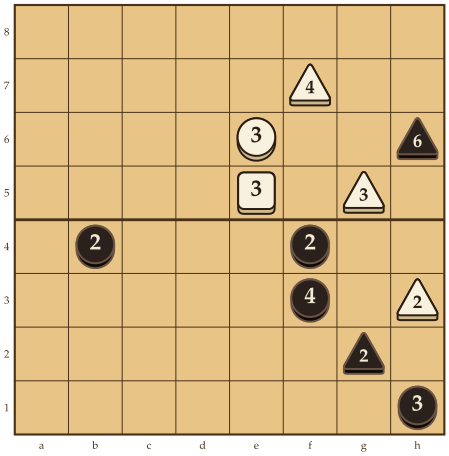
Puzzle 3



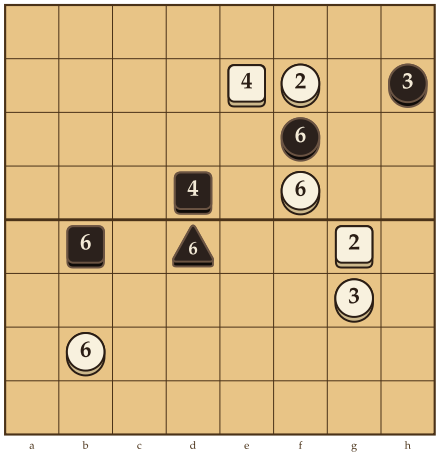
Puzzle 4



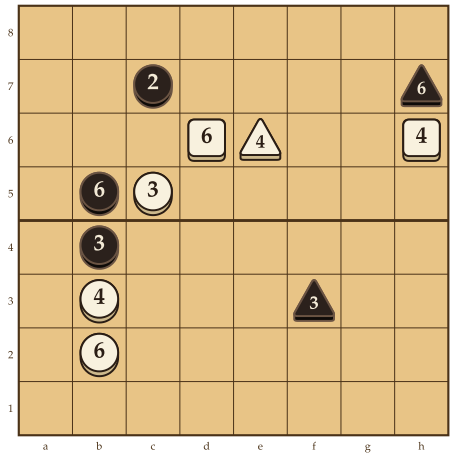
Puzzle 5



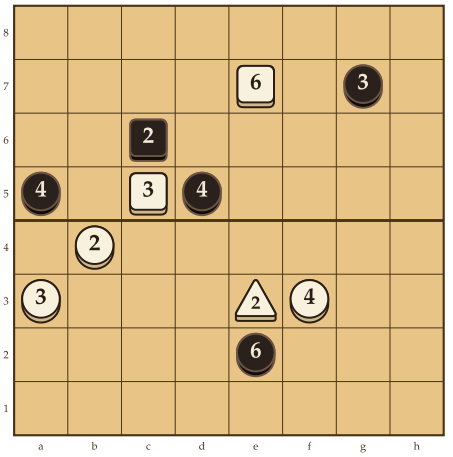
Puzzle 6



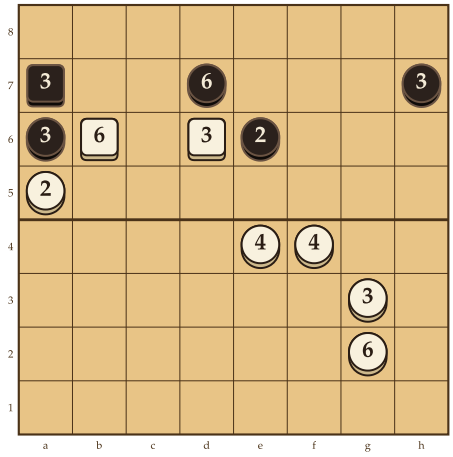
Puzzle 7



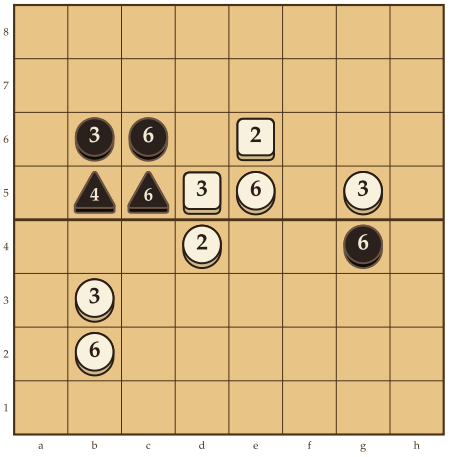
Puzzle 8



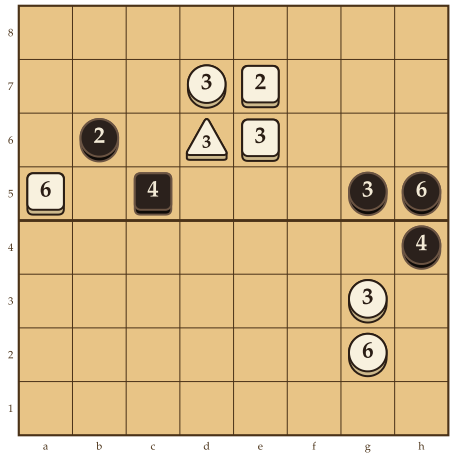
Puzzle 9



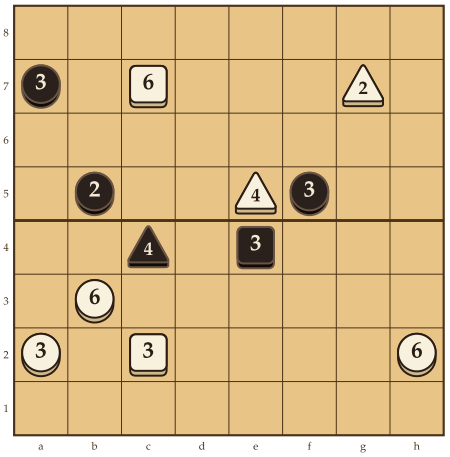
Puzzle 10



Puzzle 11



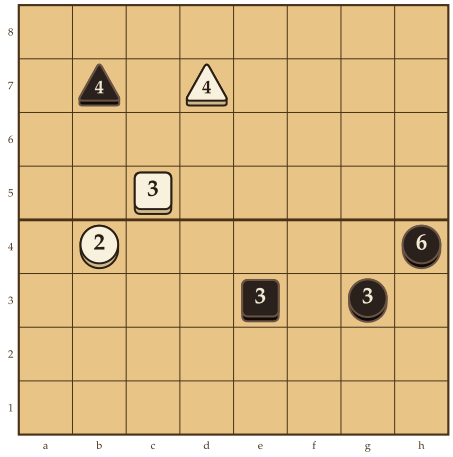
Puzzle 12



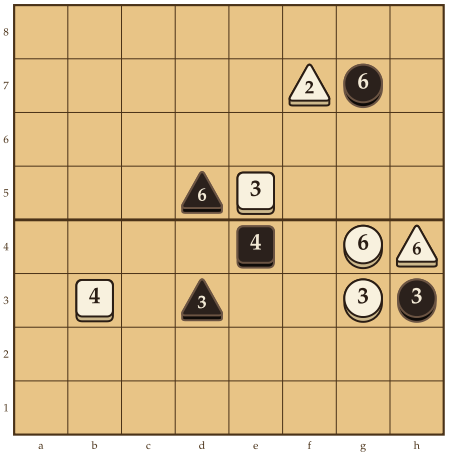
B. Win in three moves

White wins in at most three moves, whatever the defense; none of these positions can be won in two. The first move does not always threaten to win at once: sometimes you must prepare the double threat.

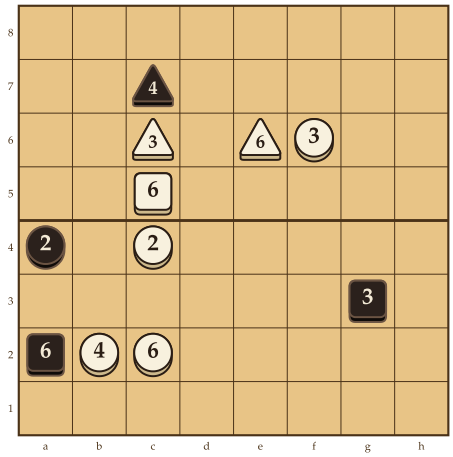
Puzzle 13



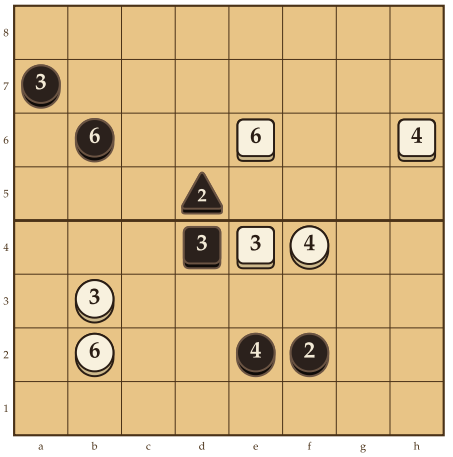
Puzzle 14



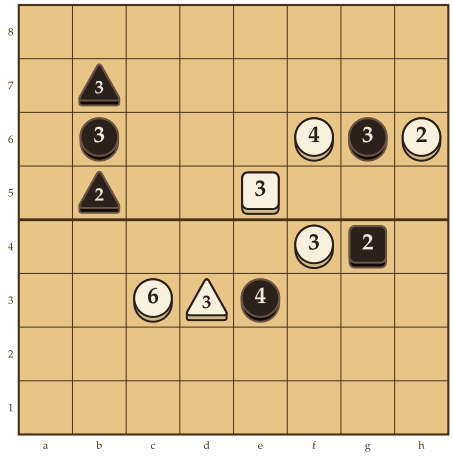
Puzzle 15



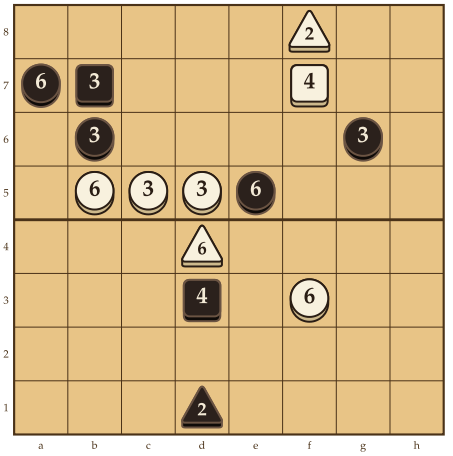
Puzzle 16



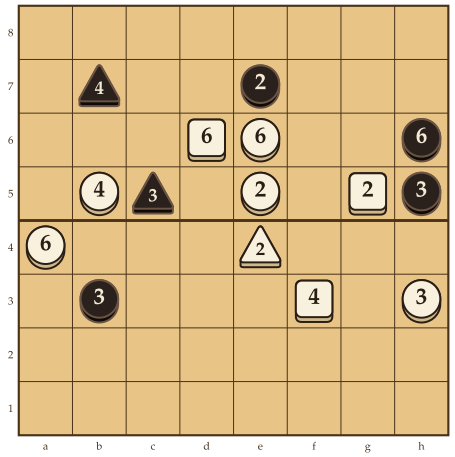
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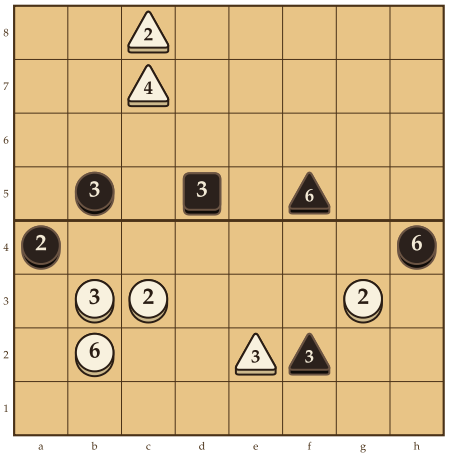
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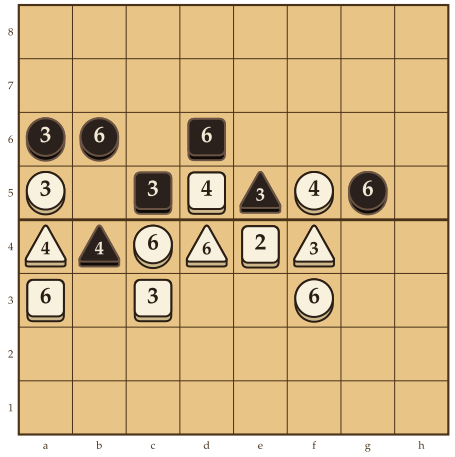
Puzzle 19



Puzzle 20



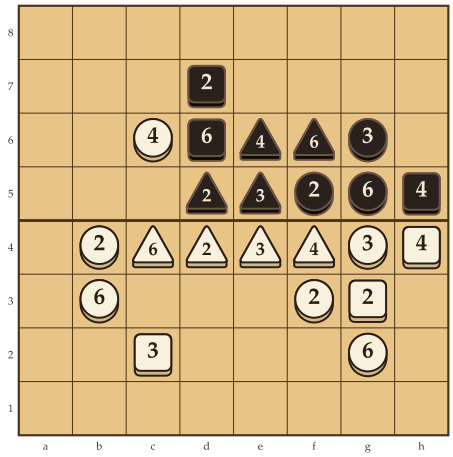
Puzzle 21



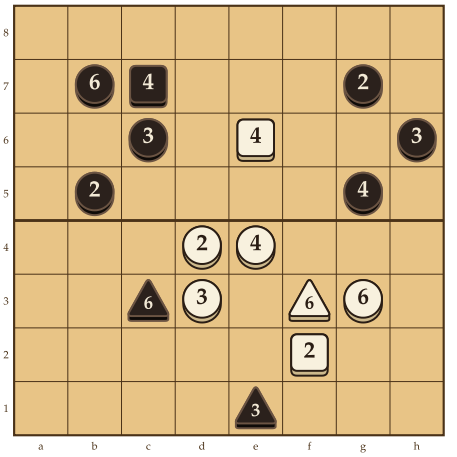
C. Win material

White wins at least the equivalent of a circle 3, whatever the defense. Each reply was checked against every defense in its group by exhaustive search; when a group contains several defenses, the sample line given is for the first one.

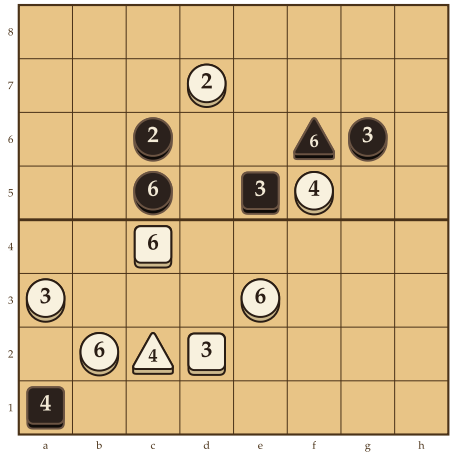
Puzzle 22



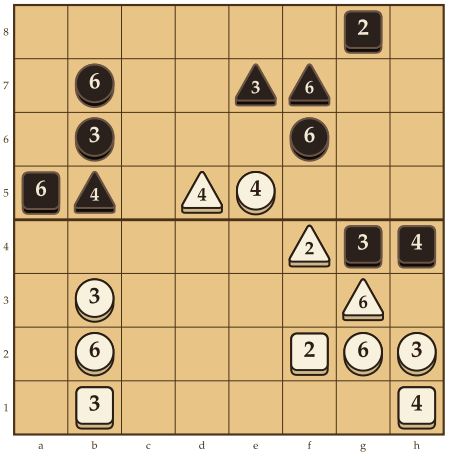
Puzzle 23



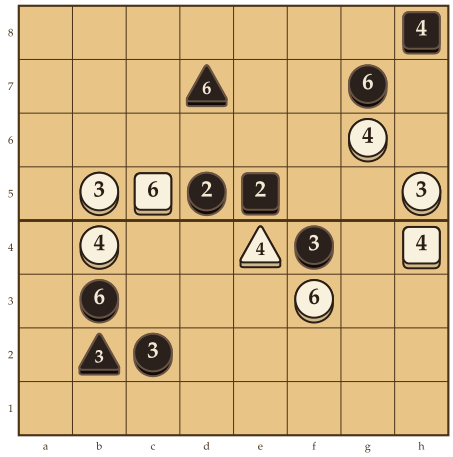
Puzzle 24



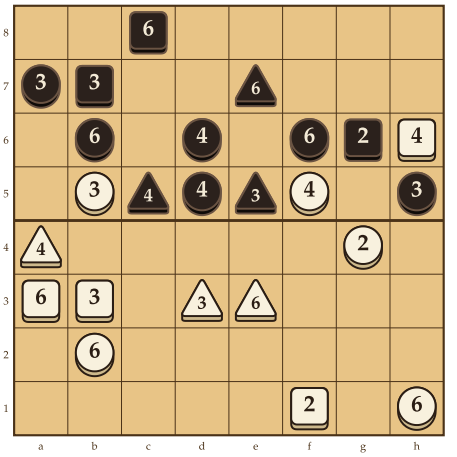
Puzzle 25



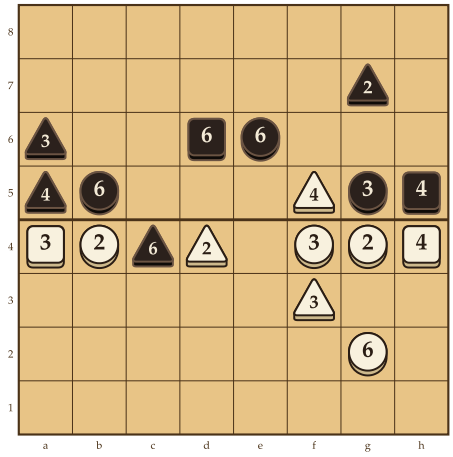
Puzzle 26



Puzzle 27



Puzzle 28



Solutions

Each solution gives every defense. Black replies that call for the same answer are grouped together; "any other reply" covers all the rest.

A. Win in two moves

1. C3 g6-h5! Threat: 2. S4 d6-g6 # (f7-g6-h5 = 6-4-3).

- 1... T4 f5-f6 2. S4 d6×f6 # (exhaustion).
- 1... T4 f5-g6 2. S4 d6×g6 # (f7-g6-h5 = 6-4-3).
- 1... T4 f5-e6 2. S4 d6×e6 # (exhaustion).

- Any other reply: 2. S4 d6-g6 #.

2. T3 g6-f6! Threat: 2. S4 g8-g6 # (f6-g6-h6 = 3-4-6).

- Against any reply: 2. S4 g8-g6 #.

3. S2 e6-g6! Threat: 2. S2 g6-h7 # (h5-h6-h7 = 6-3-2).

- Against any reply: 2. S2 g6-h7 #.

4. T6 e6-c8! Threat: 2. T6 c8-b8 # (b8-c7-d6 = 6-4-3).

- Against any reply: 2. T6 c8-b8 #.

5. T2 h3-h5! Threat: 2. T3 g5-g6 # (f7-g6-h5 = 4-3-2). Second threat: 2. T4 f7-f5 #.

- 1... T6 h6-g6 2. T4 f7-f5 # (f5-g5-h5 = 4-3-2).
- 1... T6 h6-f6 2. T3 g5-g6 (×f6) # (f7-g6-h5 = 4-3-2).
- Any other reply: 2. T3 g5-g6 #.

6. C6 f5-e6! Threat: 2. C2 f7-e8 # (e6-e7-e8 = 6-4-2). Second threat: 2. C6 e6-d7 #.

- Against any reply: 2. C2 f7-e8 #.

7. T4 e6-d5! Threat: 2. S6 d6-e5 # (c5-d5-e5 = 3-4-6).

- Against any reply: 2. S6 d6-e5 #.

8. S3 c5-d6! Threat: 2. T2 e3-c5 # (c5-d6-e7 = 2-3-6). Second threat: 2. C2 b4-c5 #.

- 1... C4 d5-c5 2. T2 e3×c5 # (c5-d6-e7 = 2-3-6).
- 1... S2 c6-c5 2. T2 e3×c5 # (c5-d6-e7 = 2-3-6).
- 1... C4 d5-d4 2. C2 b4-c5 (×d4) # (c5-d6-e7 = 2-3-6).
- 1... C4 d5-e4 (×e3) 2. C2 b4-c5 # (c5-d6-e7 = 2-3-6).
- Any other reply: 2. T2 e3-c5 #.

9. C4 e4-d5! Threat: 2. C4 d5-c6 # (b6-c6-d6 = 6-4-3).

- 1... C6 d7-c6 2. S6 b6×c6 # (exhaustion).
- Any other reply: 2. C4 d5-c6 #.

10. S2 e6-e7! Threat: 2. S3 d5-e6 # (e5-e6-e7 = 6-3-2).

- Against any reply: 2. S3 d5-e6 #.

11. S6 a5-c3! Threat: 2. S6 c3-e5 # (e5-e6-e7 = 6-3-2).

- 1... S4 c5-e5 2. S6 c3-c5 # (c5-d6-e7 = 6-3-2).
- 1... S4 c5-d4 2. S6 c3-c5 # (c5-d6-e7 = 6-3-2).
- 1... C3 g5-f4 2. S6 c3-e5 (×f4) # (e5-e6-e7 = 6-3-2).
- Any other reply: 2. S6 c3-e5 #.

12. T4 e5-f6! Threat: 2. S6 c7-e5 # (e5-f6-g7 = 6-4-2).

- 1... S3 e4-e5 2. S6 c7×e5 # (e5-f6-g7 = 6-4-2).

- 1... C3 f5-e5 2. S6 c7×e5 # (e5-f6-g7 = 6-4-2).
- Any other reply: 2. S6 c7-e5 #.

B. Win in three moves

13. S3 c5-c6! Threat: 2. C2 b4-b5 # (b5-c6-d7 = 2-3-4).

- 1... T4 b7-b5 2. C2 b4-c5! and 3. T4 d7-c7 # (c5-c6-c7 = 2-3-4) against any reply.
- Any other reply: 2. C2 b4-b5 # (b5-c6-d7 = 2-3-4).

14. C6 g4-f5! Threat: 2. S3 e5-f6 # (f5-f6-f7 = 6-3-2).

- 1... C6 g7-f6 2. S3 e5-e6! 2... C6 f6-e7 3. S3 e6-f6 # (f5-f6-f7 = 6-3-2); 2... T6 d5-d7 or T6 d5×f5 3. S4 b3-d5 # (d5-e6-f7 = 4-3-2); any other reply, 3. T2 f7-d7 # (d7-e6-f5 = 2-3-6).
- Any other reply: 2. S3 e5-f6 # (f5-f6-f7 = 6-3-2).

15. C2 c4-b5! Threat: 2. T6 e6-d7 # (b5-c6-d7 = 2-3-6).

- 1... T4 c7-d7 2. S6 c5-d6! and 3. C2 b5-b6 # (b6-c6-d6 = 2-3-6) against any reply.
- Any other reply: 2. T6 e6-d7 # (b5-c6-d7 = 2-3-6).

16. S4 h6-f6! Threat: 2. S3 e4-g6 # (e6-f6-g6 = 6-4-3).

- 1... T2 d5-f5 2. C4 f4-g4 (×f5)! and 3. S3 e4-g6 # (e6-f6-g6 = 6-4-3) against any reply.
- Any other reply: 2. S3 e4-g6 # (e6-f6-g6 = 6-4-3).

17. S3 e5-e8! Threat: 2. S3 e8×g6 # (f6-g6-h6 = 4-3-2).

- 1... C3 g6-f7 2. C3 f4-g5! 2... C3 f7-g6 3. S3 e8×g6 # (f6-g6-h6 = 4-3-2); any other reply, 3. C3 g5-g6 # (f6-g6-h6 = 4-3-2).
- 1... C3 g6-g7, C3 g6-g5, C3 g6-h7, C3 g6-h5 or C3 g6-f5 2. S3 e8-g6 # (f6-g6-h6 = 4-3-2).
- Any other reply: 2. S3 e8×g6 # (f6-g6-h6 = 4-3-2).

18. T6 d4×e5! Threat: 2. T6 e5-f6 # (f6-f7-f8 = 6-4-2).

- 1... C3 g6-f6 2. T6 e5-d6! 2... C3 f6-e7 3. T6 d6-f6 # (f6-f7-f8 = 6-4-2); 2... S3 b7-e7 3. T6 d6×f6 # (f6-f7-f8 = 6-4-2); any other reply, 3. S4 f7-e7 # (d6-e7-f8 = 6-4-2).
- Any other reply: 2. T6 e5-f6 # (f6-f7-f8 = 6-4-2).

19. S4 f3-f6! Threat: 2. T2 e4-g6 # (e6-f6-g6 = 6-4-2) or 2. S2 g5-g6 #.

- 1... C3 h5-g6 2. T2 e4-f5! 2... C3 g6-f7 3. S2 g5-g6 # (e6-f6-g6 = 6-4-2); 2... C2 e7-f7 3. S6 d6-e7 # (e7-f6-g5 = 6-4-2); any other reply, 3. C6 e6-f7 # (f5-f6-f7 = 2-4-6).
- 1... C6 h6-g6 2. C2 e5-f5! 2... C6 g6-f7 3. S2 g5-g6 # (e6-f6-g6 = 6-4-2); 2... C2 e7-f7 3. S6 d6-e7 # (e7-f6-g5 = 6-4-2); any other reply, 3. C6 e6-f7 # (f5-f6-f7 = 2-4-6).
- Any other reply: 2. T2 e4-g6 # (e6-f6-g6 = 6-4-2).

20. T4 c7-a5! Threat: 2. C6 b2-a3 (×a4) # (exhaustion).

- 1... C2 a4-b4 2. C2 c3-d4! 2... C2 b4-a4 3. C6 b2-a3 (×a4) # (exhaustion); 2... C2 b4-c4 3. T4 a5-b4 (×c4) # (exhaustion); 2... C2 b4-c5 3. T4 a5-b6 (×c5) # (exhaustion); 2... C2 b4-a3 3.

T4 a5×a3 # (exhaustion); 2... C2 b4-c3 3. T4 a5×c3 # (exhaustion); any other reply, 3. C6 b2-c3 (×b4) # (exhaustion).

- 1... C2 a4-a3 2. T4 a5×a3 # (exhaustion).
- Any other reply: 2. C6 b2-a3 (×a4) # (exhaustion).

21. T4 a4-b5! Quiet move: there is no immediate threat.

- 1... T4 b4-a4, T4 b4-b3 or T4 b4-b2 2. S6 a3×c5 # (a5-b5-c5 = 3-4-6).
- 1... S3 c5-c6, S3 c5-c7 or S3 c5-c8 2. C6 c4-c5 # (a5-b5-c5 = 3-4-6).
- 1... T3 e5-e6, T3 e5-e7 or T3 e5-g7 2. T6 d4-e3! 2... S3 c5-c6, S3 c5-c7, S3 c5-c8, S3 c5-d4 or S3 c5×e3 (×f4) 3. C6 c4-c5 # (a5-b5-c5 = 3-4-6); any other reply, 3. T6 e3×c5 # (a5-b5-c5 = 3-4-6).
- Any other reply: 2. S2 e4-c2! 2... T4 b4-a4, T4 b4-b2 or T4 b4-b3 3. S6 a3×c5 # (a5-b5-c5 = 3-4-6); any other reply, 3. S2 c2-a4 (×b4) # (exhaustion).

C. Win material

22. C2 b4-c5!

- 1... T2 d5×c5 2. C4 c6-b6 (×c5) S2 d7-e7 3. S3 c2-d2 C6 g5-h6; White wins triangle 2 and loses circle 2.
- Any other reply: 2. C2 c5×d5 C3 g6-h6 3. S3 c2-d2 T3 e5-f6 (line given after 1... T6 f6-h8); White wins triangle 2 and loses nothing.

23. S4 e6-b3!

- 1... T6 c3-b4 2. C2 d4-c4 T6 b4-a3 3. S4 b3×b5 T3 e1-f1; White wins circle 2 and loses nothing.
- 1... S4 c7-e5 (×d4) 2. C4 e4×e5 T3 e1-e3 (×d3) 3. S4 b3×b5 T6 c3-d3; White wins square 4, circle 2 and loses circle 2, circle 3.
- 1... T6 c3-c2, T6 c3-a1, T6 c3-c4, T6 c3-a5, T6 c3-c5, T6 c3-d2, T6 c3-b2 or T6 c3-c1 2. S4 b3×b5 T6 c2-b1 3. C3 d3-c4 T6 b1-c1 (line given after 1... T6 c3-c2); White wins circle 2 and loses nothing.
- Any other reply: 2. S4 b3-b2 (×c3) S4 c7-c5 3. S4 b2-c2 T3 e1-f1 (line given after 1... C3 c6-d5); White wins triangle 6 and loses nothing.

24. S3 d2-d5!

- 1... S3 e5-e6 or T6 f6-e6 2. S6 c4×c5 T6 f6-f7 3. C6 b2-a2 S4 a1-d4 (line given after 1... S3 e5-e6); White wins circle 6 and loses nothing.
- Any other reply: 2. S3 d5-e6 #; White wins the game.

25. C3 h2-h3!

- 1... S2 g8-g5 2. T2 f4×g5 S3 g4×h3 3. S3 b1-c1 S3 h3-h2; White wins square 2 and loses circle 3.
- 1... S3 g4-h5, S3 g4-g6, S3 g4-e2, S3 g4-g7, S3 g4-d1, S3 g4-d7, S3 g4×h3, S3 g4-e6, S3 g4-f5, S3 g4-f3 or S3 g4-g5 2. T2 f4×h4 T3 e7-d6 3. S3 b1-c1 S6 a5-a6 (line given after 1... S3 g4-

h5); White wins square 4 and loses nothing.

- Any other reply: 2. C3 h3×g4 T3 e6×g4 3. S3 b1-d1 T6 f7-g6 (line given after 1... T3 e7-e6); White wins square 3 and loses circle 3.

26. C3 b5-a5!

- 1... T6 d7-b5 2. S6 c5×b5 S2 e5-c3 3. S6 b5-c6 (×d5) S2 c3×f3; White wins triangle 6, circle 2 and loses circle 6.
- Any other reply: 2. C4 b4-b5 #; White wins the game.

27. C2 g4-g5!

- 1... S2 g6×g5 2. T6 e3-f4 (×g5) T6 e7-c7 3. S2 f1-g1 T4 c5-c6; White wins square 2 and loses circle 2.
- 1... T3 e5-e4, T3 e5×e3 or C4 d5-e4 2. S4 h6-g7 (×g6) T6 e7-e8 3. S2 f1-g1 T3 e4-f4 (line given after 1... T3 e5-e4); White wins square 2 and loses nothing.
- Any other reply: 2. T6 e3×e5 #; White wins the game.

28. T2 d4-d5!

- 1... S6 d6×f4 2. T4 f5-e4 (×f4) T6 c4-c3 (×b4) 3. C6 g2-h2 T6 c3-e3; White wins square 6 and loses circle 3, circle 2.
- 1... T2 g7-e5, S6 d6-e5 or C6 e6-e5 2. T4 f5-d3 T2 e5-c3 (×b4) 3. S3 a4×c4 T2 c3-a3 (line given after 1... T2 g7-e5); White wins triangle 6 and loses circle 2.
- Any other reply: 2. C3 f4-e5 #; White wins the game.

Appendix

Methods and statistics

The figures in this manual come from experiments run with two game engines written for the reference rules. Every game was replayed by a second, independent implementation of the rules, without a single discrepancy.

The Engines

Engine	Search	Level	Use
Medium	4 plies, evaluation learned from 73,000 positions	club	500-game corpus, first experiments
Strong	6 plies, same evaluation, null move and transposition table	wins 39 games out of 40 against Medium	confirmation of every principle

The Experiments

Question	Method	Size
Piece values	regression of the result on the remaining material, with a scarcity term per number	575 games, 4,518 positions
Weight of each principle	full engine against the same engine with one term removed from its evaluation	40 games per principle
Endgames	randomly drawn positions, each side in its own half, played twice with colors reversed	120 games per situation
Openings	first four moves imposed on one side, then free play	40 games per match at the strong level
Strategic factors	regression of the result on positional features, at equal material, by phase	102 strong games, 1,750 positions
Anatomy of the triumph	record of the winning line and the moves leading up to it	83 triumphs from strong games
Puzzles	positions where only one move wins, rechecked by exhaustive 6-ply search	28 selected, with the full tree of their defenses

Margins of Error

A score measured over 40 games is accurate to within about 15 points; over 120 games, to within about 9 points. The principles retained show differences well above these margins. The openings are the least certain part of the manual.

Limitations

The strong engine remains a good player, not a master. It calculates six plies ahead, and its evaluation is linear. The principles in this manual will have to be confirmed, refined or corrected by strong human players: that is the fate of every new theory.